

Project Development

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

This phase describes the complete development process of the ServiceNow ACL project. The project was developed in a ServiceNow Personal Developer Instance by creating users, roles, a custom table, and Access Control Lists for record-level security.

2. Development Environment

Platform: ServiceNow Personal Developer Instance (Australia)

Browser: Google Chrome / Microsoft Edge

Technology: ServiceNow Access Control Lists (ACL), User Administration, System Definition Tables

3. User and Role Development

Created one test user named EEE User. Created four custom roles (bb1, bb2, bb3, bb4) and assigned all roles to the EEE User for testing different permission levels.

4. Custom Table Development

Created a custom table named Institution Details (u_institution_details). Added the following fields:

- Student Roll Number (Auto Number)
- Student Name (Reference - User)
- Faculty Name (Reference - User)
- Branch (Choice: EEE, ECE, CSE)
- Email (String)
- Phone Number (String)
- Description (Multi-line String)

5. ACL Implementation

ACL Type	Role	Purpose
READ	bb1	View only EEE branch records.
CREATE	bb2	Allow users to create new records.
WRITE	bb3	Allow users to edit EEE records.
DELETE	bb4	Allow users to delete EEE records.

6. Script Logic

The READ ACL contains a script that checks whether the user has the Admin role or the bb1 role. Admin users receive unrestricted access, while users with the bb1 role can access only EEE branch records.

7. Verification During Development

- Impersonated EEE User and verified only EEE records were visible.
- Impersonated Admin user and verified all records were visible.
- Verified CREATE, WRITE, and DELETE permissions using bb2, bb3, and bb4 roles.

8. Expected Development Output

The project successfully implemented secure record-level access using ServiceNow ACLs. Different users received permissions according to their assigned roles, ensuring secure access to Institution Details records.

The top screenshot shows the 'Access Control - New Record' form in ServiceNow. It includes a warning message: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' The form fields are as follows:

- * Type: record
- * Operation: read
- Application: Global
- Active: ☒
- Advanced: ☐
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: (empty)

The bottom screenshot shows the 'Access Control - u_institution_details' record. The fields are:

- * Type: record
- * Operation: create
- Application: Global
- Active: ☒
- Advanced: ☐
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: Allow create for records in u_institution_details, for users with role bb2.

Below the form, the 'Conditions' section is expanded, showing a message: 'Access Control Rules have two decision types, and these types will behave differently depending on conditions. 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access. 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access. [More Info](#)'

At the bottom, the 'Requires role' section shows a pagination bar: '1 to 1 of 1'.

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AllFavoritesHistoryAdmin

Access Controls

Search

Access Controlu_institution_details

UpdateDelete

* Typerecord

* Operationwrite

Admin overrides☒

Decision TypeAllow If

Protection policy-- None --

* NameInstitution Details [u_institution_details]

DescriptionAllow write for records in u_institution_details, for users with role bb3.

ApplicationGlobal

Active☒

Advanced☐

Conditions

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AllFavoritesHistoryAdmin

Access Controls

Search

Access Controlu_institution_details

UpdateDelete

* Typerecord

* Operationdelete

Admin overrides☒

Decision TypeAllow If

Protection policy-- None --

* NameInstitution Details [u_institution_details]

DescriptionAllow delete for records in u_institution_details, for users with role bb4.

ApplicationGlobal

Active☒

Advanced☐

Conditions